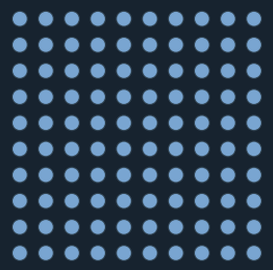
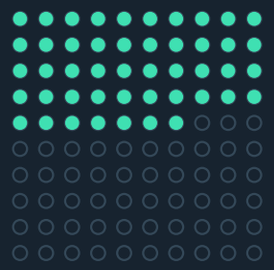
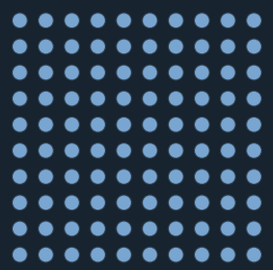
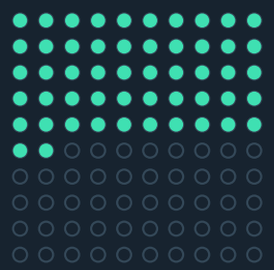
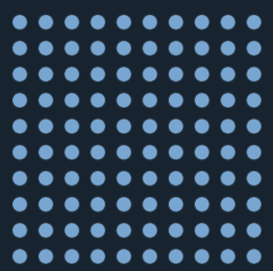
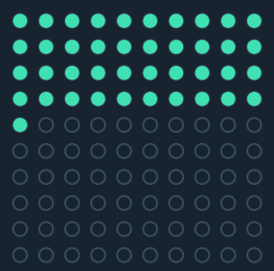
BEFORE → AFTER

Same numerical outputs. Fewer instructions.

1.94×

geometric mean improvement

	BEFORE · ESP-NN C	AFTER · TUNED RUST	INSTRUCTIONS SAVED
3×3 convolution 16×16×16 → 16×16×32	 8,546,046	 4,032,868	**52.8%** 2.12× improvement −4,513,178 instructions
1×1 convolution 16×16×32 → 16×16×32	 2,129,234	 1,105,945	**48.1%** 1.93× improvement −1,023,289 instructions
Depthwise 3×3 32×32×16	 1,955,993	 1,372,290	**29.8%** 1.43× improvement −583,703 instructions
Fully connected 256 → 128	 302,154	 124,485	**58.8%** 2.43× improvement −177,669 instructions

Each matrix = 100% of that operator’s C baseline. Filled dots = instructions still used.

Dots are rounded to 1%; numeric labels are exact. QEMU guest instructions ≠ hardware cycles or elapsed time.